BLOCKCHAIN Experiment No: 8

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D20A/24

Aim : Configuring a Cryptocurrency Wallet using Metamask and integrating it with a DApp

Theory :

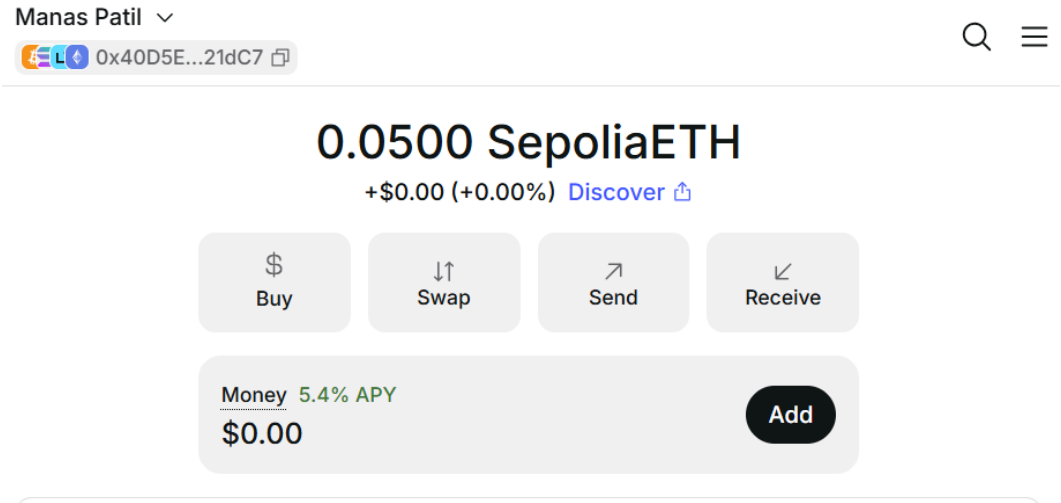
MetaMask is a cryptocurrency wallet that allows users to securely manage Ethereum accounts, store digital assets, and interact with blockchain networks. It acts as a bridge between users and decentralized applications (DApps), enabling them to connect their wallets and authorize blockchain transactions.

A Decentralized Application (DApp) is an application that interacts with blockchain-based smart contracts instead of depending entirely on centralized servers. Smart contracts are self-executing programs deployed on a blockchain that perform operations according to predefined rules.

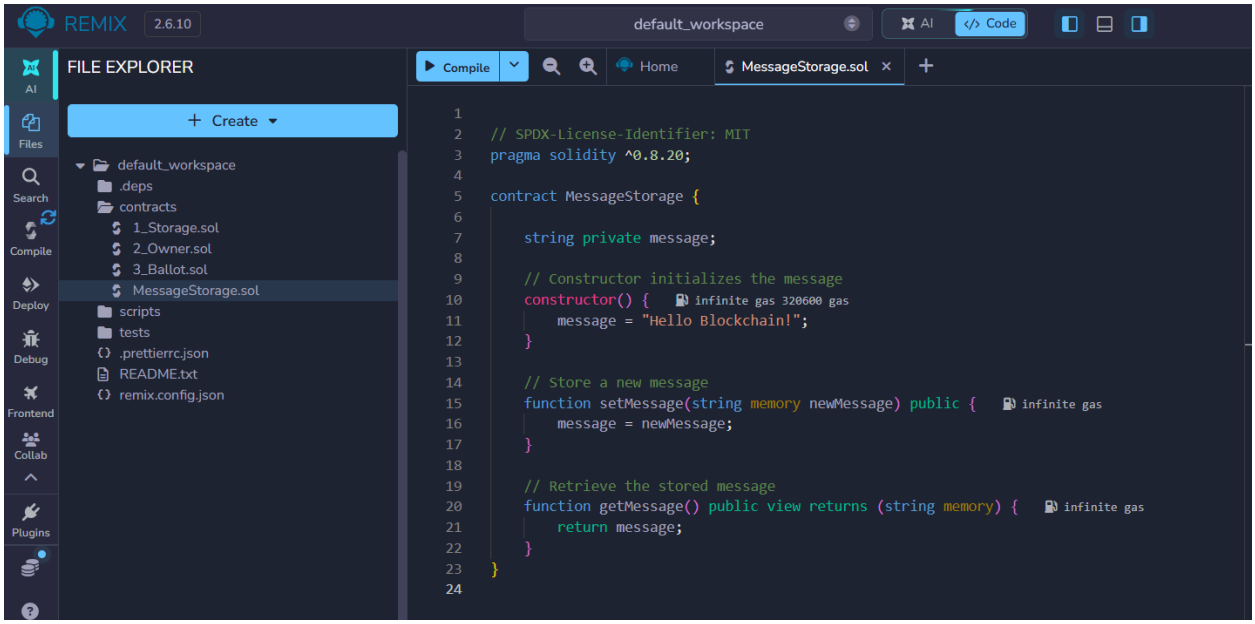
In this experiment, MetaMask was configured with the Ethereum Sepolia test network, and a cryptocurrency wallet was created. A smart contract was developed using Solidity and deployed through Remix IDE. A DApp interface was created using HTML, CSS, JavaScript, and Ethers.js to connect MetaMask with the deployed smart contract. The application allowed users to retrieve and update messages stored on the blockchain. The transactions were authorized through MetaMask and verified using Sepolia Etherscan.

Procedure :

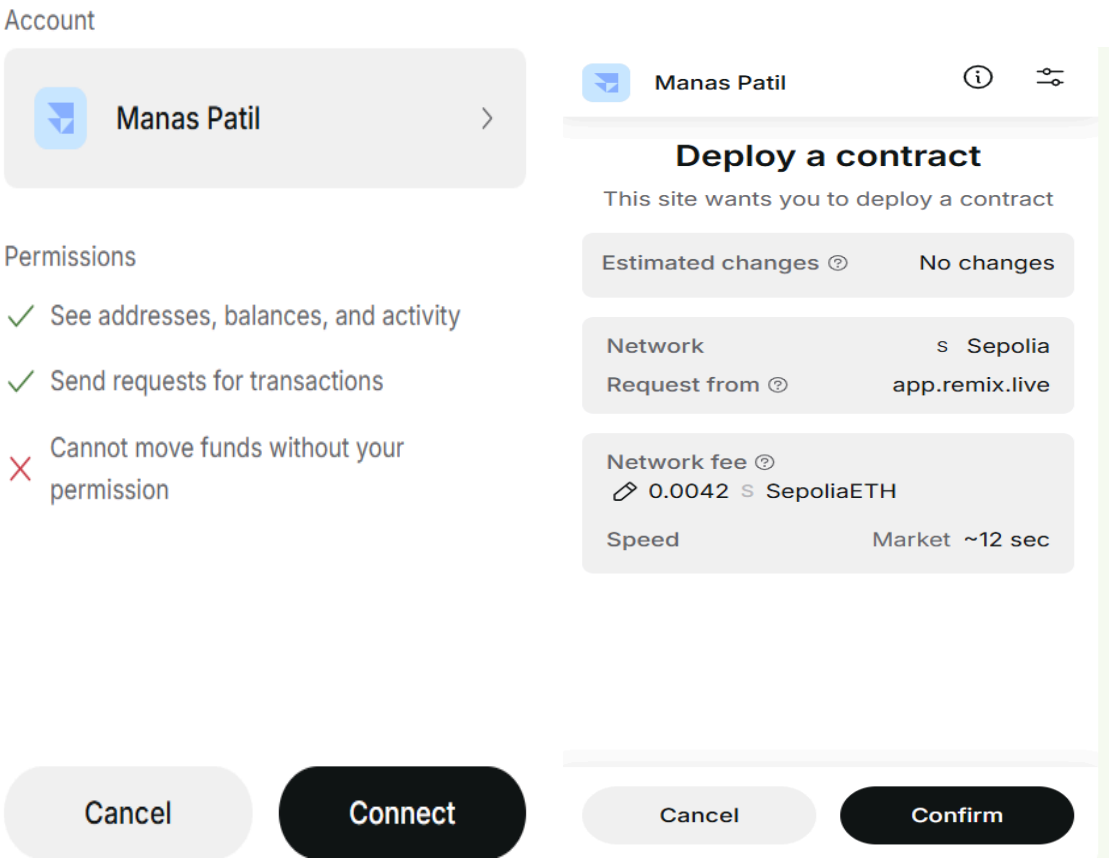
Step 1: Getting Test ETH into MetaMask



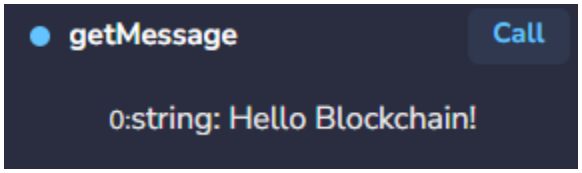
Step 2 : Creating a Smart Contract Using Remix IDE



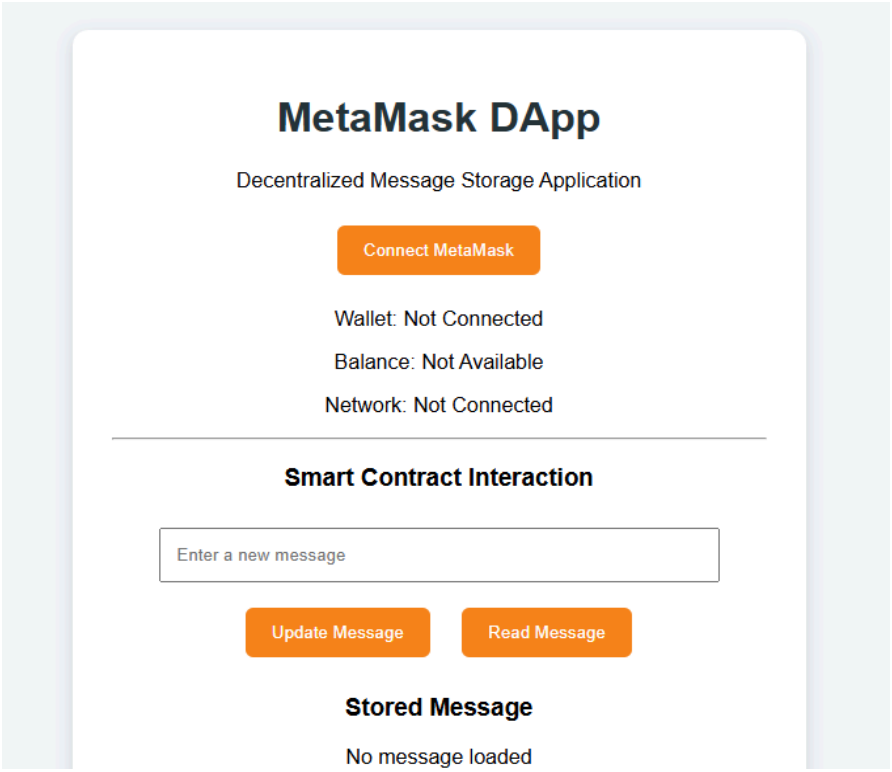
Step 3 : Deploying the Smart Contract Using MetaMask :



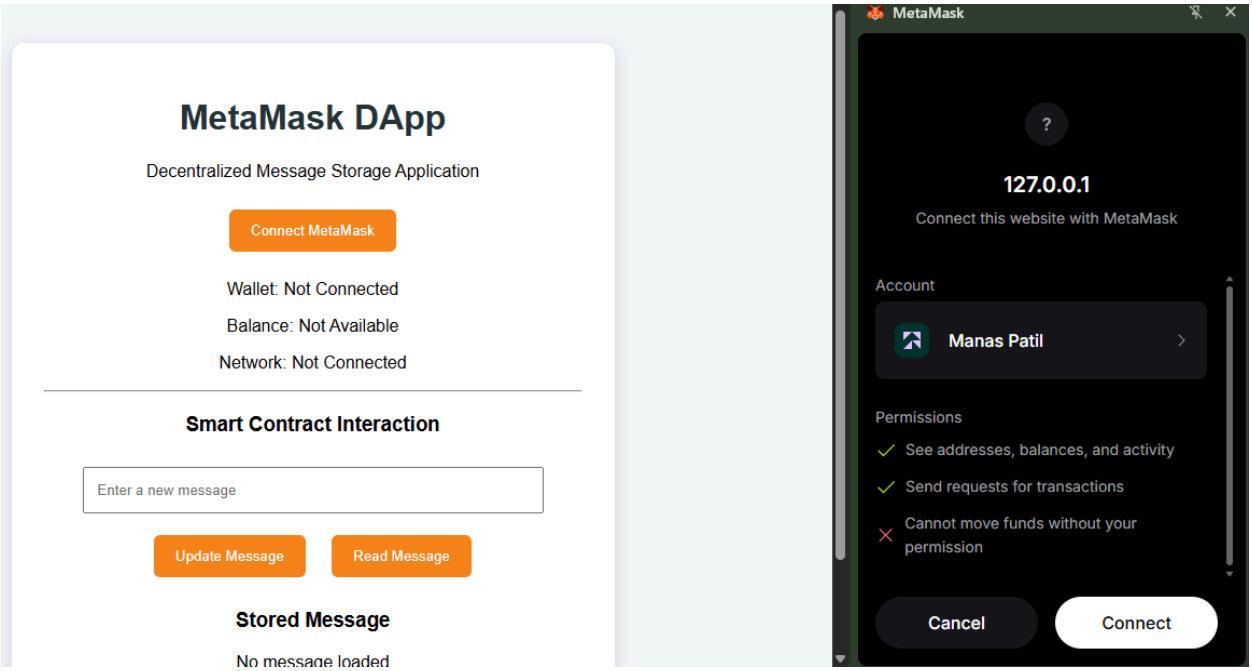
Verify successful deployment



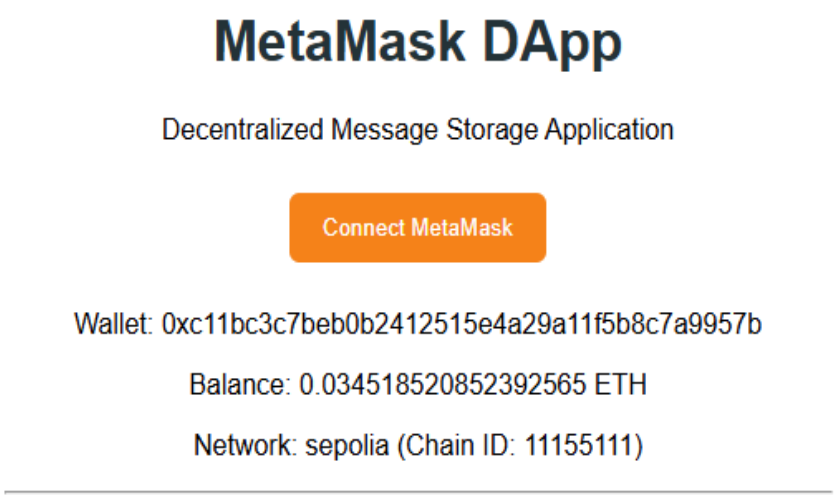
Step 4 : Developing the DApp Frontend



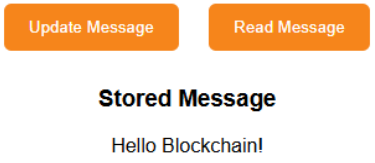
Step 5: Connecting MetaMask with the DApp



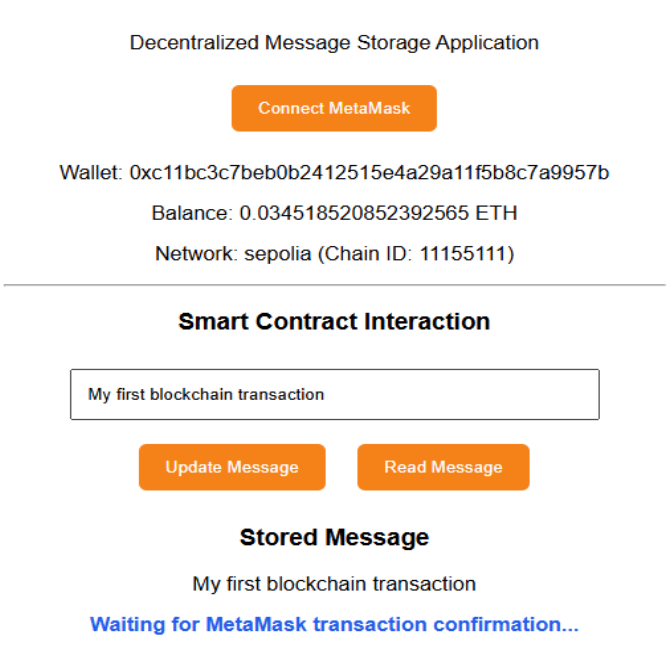
After connecting your DApp should now display:

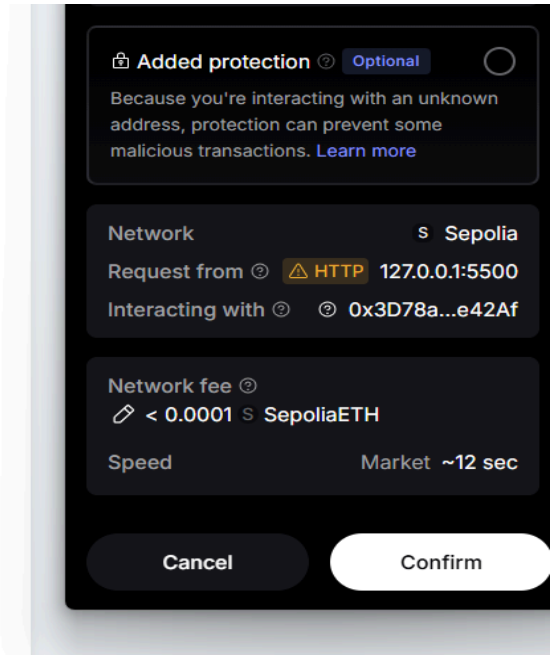


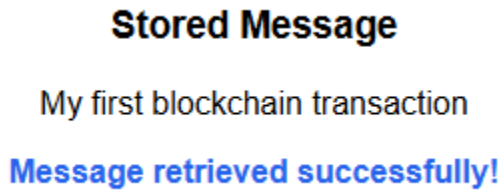
Step 6 : Read the message from the smart contract



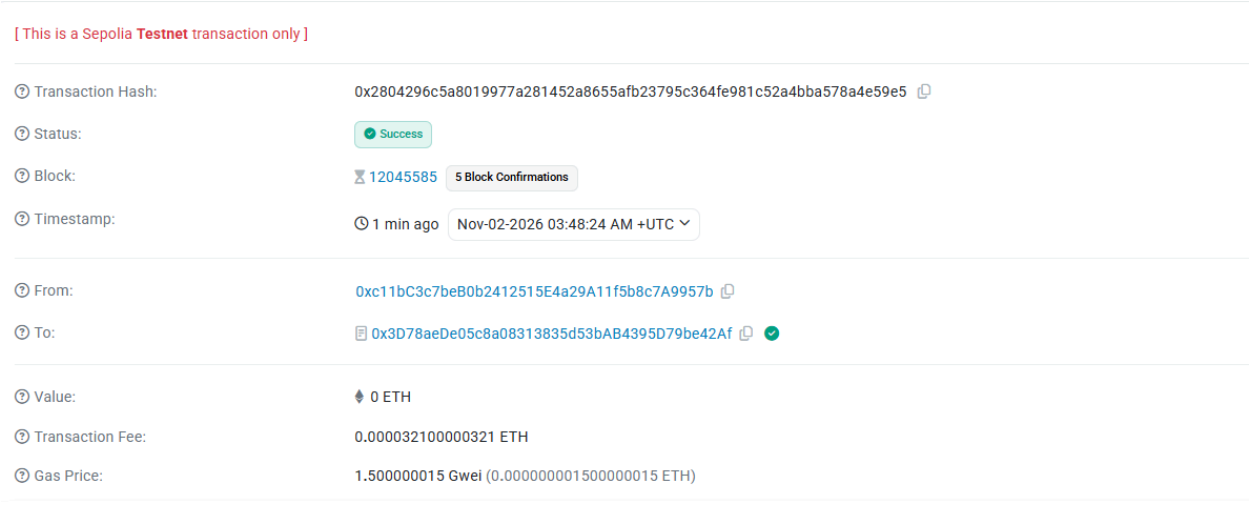
Step 7 : Update the message through the DApp



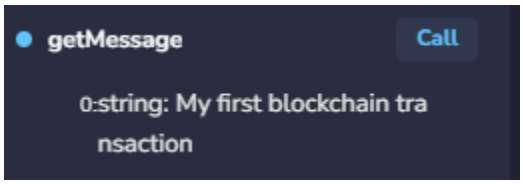




Step 8 : Verifying Blockchain Transactions



Step 9 : Verify the updated message in Remix



Conclusion :

Thus, a cryptocurrency wallet was successfully configured using MetaMask and integrated with a Decentralized Application. A Solidity smart contract was deployed on the Sepolia test network, and blockchain transactions were performed through the DApp. The transaction hash was verified using Etherscan, demonstrating how MetaMask enables secure wallet connectivity and interaction with blockchain-based applications.